University of Dhaka

Assignment - 2

Course Title: Advanced Econometrics-II
Course Code: 408

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Question:

A data analyst estimated an ARMA (2,1) model for changes in the clothing industry's profits. However, the true model is actually an ARMA (2,2). Use simulations to show that the data analyst will estimate a biased model.

Solution:

Let's assume an ARMA (2,2) model is,

$$y_t = 0.4y_{t-1} + 0.3y_{t-2} + \varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-2}$$

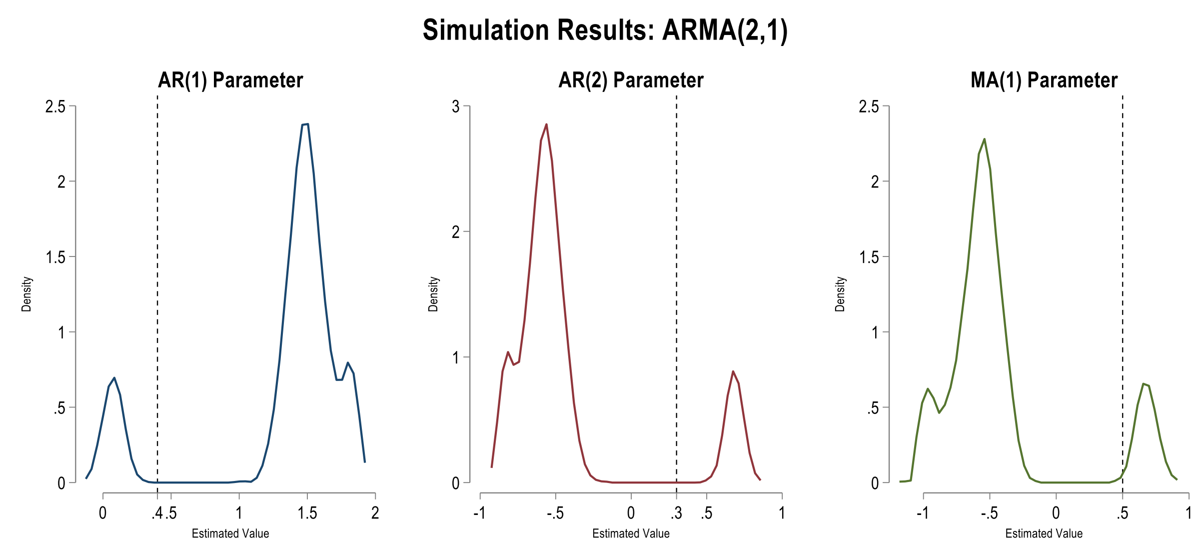
Where the coefficient of AR (1) = 0.4

coefficient of AR (2) = 0.3

coefficient of MA (1) = 0.5

coefficient of MA (2) = 0.3

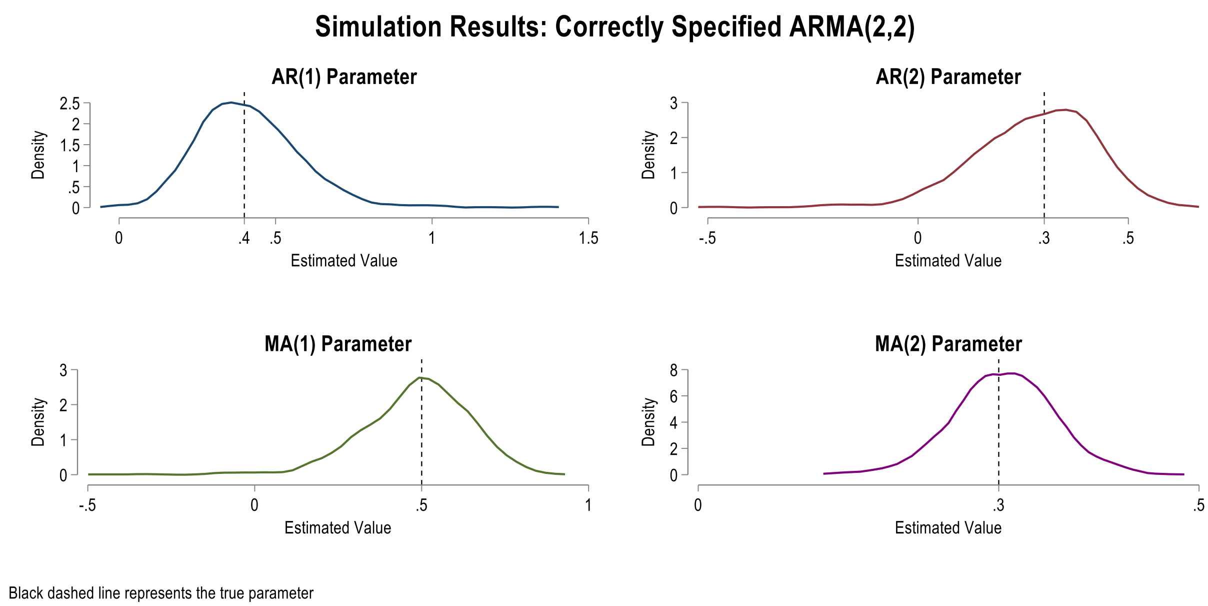
Now if a data analyst estimated an ARMA (2,1) model then the estimated coefficients will be biased downward or upward. To show this we will run a simulation with 1000 samples. With each sample we will estimate the coefficient, and then we will compare the expected value or mean of coefficients with the true parameters.



Black dashed line represents the true parameter

Coefficients	True value	Estimated Value	Biased
AR (1)	0.4	1.32	Upward by 0.92 points
AR (2)	0.3	-0.42	Downward by 0.72 points
MA (1)	0.5	-0.43	Downward by 0.93 points

From the diagram we can see that the estimated values exhibit multimodal distribution and does not converge to the single true parameters. The AR (1) parameter is estimated biased upward and both the AR (2) and MA (1) parameters are estimated biased downward.



Now if the data analyst had correctly specified the ARMA (2,2) model then the simulation with 1000 samples would yield similar estimated values close to true parameters and their distribution would follow the same true parameters which we can see from diagram for correctly specified ARMA (2,2) model.

In conclusion, a miss specified ARMA (2,1) model would yield biased coefficients for each value when the true model is ARMA (2,2).

```
----------------------------------------------------------------------------------------
> ------------------------------------------------------------------
      name:  <unnamed>
       log:  F:\ECON 408\Practice do file\Homework-2.log
  log type:  text
 opened on:  14 May 2026, 19:45:30

.
. * (1) Biased ARMA(2,1) Model Simulation
. *--------------------------------------------------------------------------------
.
. clear all

. set seed 1000

.
. capture postclose buffer

. postfile buffer beta_L1 beta_L2 alpha_L1 using arma22_misspecified.dta, replace

.
. forvalues i = 1/1000 {
  2.     qui drop _all
  3.         qui set obs 500
  4.         qui gen time = _n
  5.         qui tsset time
  6.
.         qui gen double e = rnormal()
  7.         qui gen double y = e in 1
  8.         qui replace y = 0.4*L1.y + e + 0.5*L1.e in 2
  9.         qui replace y = 0.4*L1.y + 0.3*L2.y + e + 0.5*L1.e + 0.3*L2.e in 3/L
 10.         qui arima y, ar(1/2) ma(1)
 11.         post buffer (_b[ARMA:L1.ar]) (_b[ARMA:L2.ar]) (_b[ARMA:L1.ma])
 12.
. }

.
. postclose buffer

. use arma22_misspecified.dta,clear

.
. sum beta_L1 beta_L2 alpha_L1

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     beta_L1 |      1,000    1.324907    .5218545  -.0861322   1.883718
     beta_L2 |      1,000   -.4230766    .4589573  -.8911305   .8230543
    alpha_L1 |      1,000   -.4310591    .4753887   -1.13755     .86856

.
. set scheme white_tableau

. graph set window fontface "Arial Narrow"

.
. * AR(1) Plot
. kdensity beta_L1, lcolor(navy) lpattern(solid) lwidth(medthick) ///
>          xline(0.4, lcolor(black) lpattern(shortdash)) ///
>                  name(g1, replace) title("{bf:AR(1) Parameter}") ///
>                  xtitle("Estimated Value") ///
>                  xlabel(0 0.4 0.5 1 1.5 2 ,labsize(medium) nogrid) ///
>                  ylabel(, labsize(medium) nogrid) note(" ")
```

```
.
. * AR(2) Plot
. kdensity beta_L2, lcolor(maroon) lpattern(solid) lwidth(medthick) ///
>          xline(0.3, lcolor(black) lpattern(shortdash)) ///
>                  name(g2, replace) title("{bf:AR(2) Parameter}") ///
>                  xtitle("Estimated Value") ///
>                  xlabel(-1 -0.5 0 0.3 0.5 1 ,labsize(medium) nogrid) ///
>                  ylabel(, labsize(medium) nogrid) note(" ")

.
. * MA(1) Plot
. kdensity alpha_L1, lcolor(forest_green) lpattern(solid) lwidth(medthick) ///
>          xline(0.5, lcolor(black) lpattern(shortdash)) ///
>                  name(g3, replace) title("{bf:MA(1) Parameter}") ///
>                  xtitle("Estimated Value") xlabel( ,labsize(medium) nogrid) ///
>                  note(" ") ylabel(, labsize(medium) nogrid)

.
. * Combine Graph
. graph combine g1 g2 g3, ///
>       rows(1) title("{bf:Simulation Results: ARMA(2,1)}") ///
>           xsize(8) ysize(4) ///
>           note("Black dashed line represents the true parameter") ///
>           name(Biased, replace)

.
. graph export Biased_arma.emf, name(Biased) replace
file F:\ECON 408\Practice do file\Biased_arma.emf saved as Enhanced Metafile format

.
.
.
. * (2) Unbiased Correct ARMA(2,2) Model Simulation
. *-------------------------------------------------------------------------------
.
. clear all

. set seed 1000

.
. capture postclose buffer

. postfile buffer beta_L1 beta_L2 alpha_L1 alpha_L2 using arma22_correct_model.dta, re
> place

.
. forvalues i = 1/1000 {
  2.     qui drop _all
  3.         qui set obs 500
  4.         qui gen time = _n
  5.         qui tsset time
  6.
.         qui gen double e = rnormal()
  7.         qui gen double y = e in 1
  8.         qui replace y = 0.4*L1.y + e + 0.5*L1.e in 2
  9.         qui replace y = 0.4*L1.y + 0.3*L2.y + e + 0.5*L1.e + 0.3*L2.e in 3/L
 10.         qui arima y, ar(1/2) ma(1/2)
 11.         post buffer (_b[ARMA:L.ar]) (_b[ARMA:L2.ar]) (_b[ARMA:L.ma]) (_b[ARMA:L2.
> ma])
 12.
```

```
. }

.
. postclose buffer

. use arma22_correct_model.dta,clear

.
. sum beta_L1 beta_L2 alpha_L1 alpha_L2

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     beta_L1 |      1,000    .4144489    .1664455  -.0255998   1.370499
     beta_L2 |      1,000    .2762445    .1460104  -.4907512   .6360206
    alpha_L1 |      1,000    .4834226    .1622318  -.4647181   .8953954
    alpha_L2 |      1,000    .3050435    .0505966   .1358287   .4744434

.
. set scheme white_tableau

. graph set window fontface "Arial Narrow"

.
.
. * AR(1) Plot
. kdensity beta_L1, lcolor(navy) lpattern(solid) lwidth(medthick) ///
>     xline(0.4, lcolor(black) lpattern(shortdash)) ///
>     name(g1, replace) title("{bf:AR(1) Parameter}") ///
>     xtitle("Estimated Value" ,size(medium)) ytitle(, size(medium)) note(" ") ///
>     xlabel(0 0.4 0.5 1 1.5, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

.
. * AR(2) Plot
. kdensity beta_L2, lcolor(maroon) lpattern(solid) lwidth(medthick) ///
>     xline(0.3, lcolor(black) lpattern(shortdash)) ///
>     name(g2, replace) title("{bf:AR(2) Parameter}") ///
>     xtitle("Estimated Value", size(medium)) ytitle(,size(medium)) note(" ") ///
>     xlabel(-0.5 0 0.3 0.5, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

.
. * MA(1) Plot
. kdensity alpha_L1, lcolor(forest_green) lpattern(solid) lwidth(medthick) ///
>     xline(0.5, lcolor(black) lpattern(shortdash)) ///
>     name(g3, replace) title("{bf:MA(1) Parameter}") ///
>     xtitle("Estimated Value", size(medium)) ytitle(,size(medium)) note(" ") ///
>     xlabel(-0.5 0 0.5 1, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

.
. * MA(2) Plot
. kdensity alpha_L2, lcolor(purple) lpattern(solid) lwidth(medthick) ///
>     xline(0.3, lcolor(black) lpattern(shortdash)) ///
>     name(g4, replace) title("{bf:MA(2) Parameter}") ///
>     xtitle("Estimated Value",size(medium)) ytitle(,size(medium)) note(" ") ///
>     xlabel(0 0.3 0.5, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

.
. * Combine into 2x2 Grid
. graph combine g1 g2 g3 g4, ///
>     rows(2) cols(2) ///
>     title("{bf:Simulation Results: Correctly Specified ARMA(2,2)}") ///
>     note("Black dashed line represents the true parameter", size(small)) ///
>     xsize(8) ysize(4) name(CorrectModel, replace)
```

```
.
.
. graph export ARMA_Correct.emf, name(CorrectModel) replace
file F:\ECON 408\Practice do file\ARMA_Correct.emf saved as Enhanced Metafile format

.
.
.
. log close
      name:  <unnamed>
       log:  F:\ECON 408\Practice do file\Homework-2.log
  log type:  text
 closed on:  14 May 2026, 19:55:07
-------------------------------------------------------------------------------------
> -----------------------------------------------------------------
```